

A noncommutative compact group with one-sided Kwapień autoduality

Candidate full-solution packet

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Status: candidate full proof likely valid for both final questions in arXiv:math/0312245. The first answer is affirmative: $\mathbb{T} \times SU(2)$ has the requested autoduality and unbounded representation dimensions. The second answer is negative: unbounded dimensions alone do not make Banach and complete Γ -type or cotype equivalent.

1 Source questions

García-Cuerva and Parcet define Γ -type and Γ -cotype for the quantized orthonormal system formed by the irreducible unitary representations of a compact group G [1]. At the end of the paper they ask:

1. Does there exist a noncommutative compact group G such that either Γ -type 2 or Γ -cotype 2 forces an operator space to be completely isomorphic to an OH space?
2. If the irreducible dimensions $\mathbf{d}_\Gamma$ are unbounded, must Banach Γ -type 2 coincide with Γ -type 2, and similarly for cotype?

The paper recalls immediately beforehand that the one-sided autoduality in the first question holds for the torus $\mathbb{T}$ and the Cantor group.

2 Proof Intuition

One-sided autoduality passes upward through compact-group quotients. Indeed, if $G \twoheadrightarrow Q$, every irreducible representation of Q pulls back to an irreducible representation of G , and the corresponding Fourier subsystem on G is exactly the Fourier system on Q . Thus type or cotype for the full dual of G implies the same property for the dual of Q .

This makes $G = \mathbb{T} \times SU(2)$ decisive. Its torus quotient supplies autoduality, while its $SU(2)$ factor makes the group noncommutative and gives irreducible representations of arbitrarily large dimension. The same example then separates Banach from complete type and cotype using the operator spaces R_p already analyzed in the source.

3 Quotient inheritance

Lemma 1 (Autoduality lifts through quotients). *Let $\pi : G \rightarrow Q$ be a continuous surjective homomorphism of compact groups. Let E be an operator space. If E has $\widehat{G}$ -type 2, then it has $\widehat{Q}$ -type 2; if it has $\widehat{G}$ -cotype 2, then it has $\widehat{Q}$ -cotype 2. The corresponding constants do not increase. Consequently, if Q has one-sided Kwapień autoduality, then so does G .*

Proof. The map

$$\widehat{Q} \longrightarrow \widehat{G}, \quad [\sigma] \longmapsto [\sigma \circ \pi]$$

is injective, preserves irreducibility, and preserves representation dimensions. Its image is the family of irreducible representations of G that are trivial on $\ker \pi$.

Normalized Haar measure on G pushes forward to normalized Haar measure on Q . Hence pullback $f \mapsto f \circ \pi$ identifies the E -valued L^2 spaces on Q completely isometrically with the subspace of functions on G that are constant on cosets of $\ker \pi$. Under this identification, the Fourier and inverse Fourier transforms for Q are precisely the restrictions of the transforms for G to the subsystem $\{\sigma \circ \pi : \sigma \in \widehat{Q}\}$.

The definitions of Γ -type and Γ -cotype take suprema over all finite subsets of the dual system. Restricting those suprema to this subsystem proves both asserted inequalities. If either property for $\widehat{Q}$ forces $E \simeq_{cb} \text{OH}$, the same is therefore true for $\widehat{G}$. $\square$

Theorem 2 (Affirmative answer to the first question). *The compact group*

$$G = \mathbb{T} \times SU(2)$$

is noncommutative, has unbounded irreducible dimensions, and satisfies: if an operator space E has $\widehat{G}$ -type 2 or $\widehat{G}$ -cotype 2, then E is completely isomorphic to an OH operator space.

Proof. Projection onto the first factor is a compact-group quotient $G \twoheadrightarrow \mathbb{T}$. The source records the one-sided autoduality of the torus in exactly the operator-space sense used here. Lemma 1 therefore proves the conclusion.

The group is noncommutative because $SU(2)$ is. Its irreducible representations include $1 \otimes \rho_m$, where ρ_m is the $(m+1)$ -dimensional irreducible representation of $SU(2)$. Thus $\mathbf{d}_{\widehat{G}}$ is unbounded. $\square$

The same proof gives a family rather than an isolated example: every noncommutative compact group admitting the torus or the Cantor group as a quotient has the requested autoduality.

4 Banach type and complete type are not equivalent

Theorem 3 (Negative answer to the second question). *Unboundedness of $\mathbf{d}_{\Gamma}$ does not imply equivalence between Banach Γ -type 2 and Γ -type 2, nor between the corresponding cotype notions. This already fails for $\Gamma = \widehat{\mathbb{T} \times SU(2)}$.*

Proof. Let R_p be Pisier's operator-space structure on the closed Rademacher span used in the source. Point (a₁) of the final example in [1] states that for every uniformly bounded quantized orthonormal system with unbounded dimensions, R_p has Banach type 2 whenever $2 \leq p < \infty$. Choose $p > 2$ and use the complete dual system of $G = \mathbb{T} \times SU(2)$. If R_p also had $\widehat{G}$ -type 2, Theorem 2 would force $R_p \simeq_{cb} \text{OH}$. The source explicitly recalls that this is impossible unless $p = 2$. Thus Banach $\widehat{G}$ -type 2 does not imply complete $\widehat{G}$ -type 2.

Similarly, point (a₂) states that R_p has Banach cotype 2 for $1 \leq p \leq 2$. Choosing $p < 2$, complete $\widehat{G}$ -cotype 2 would again imply $R_p \simeq_{cb} \text{OH}$, a contradiction. This separates the two cotype notions as well. $\square$

5 Scope and novelty

The construction answers both questions exactly as written and even keeps the surrounding unbounded-dimension hypothesis. It also exposes a natural stronger surviving problem: require the compact noncommutative group to have no torus or Cantor quotient (for example, finite abelianization), or ask for autoduality generated intrinsically by its higher-dimensional irreducibles. Nothing here settles that strengthened version, such as the case of $SU(2)$ alone.

A bounded search through 11 August 2026 covered the run’s result and attempt indexes and searches for the exact title, the two final questions, “autoduality,” quantized type/cotype, and later papers on sharp type and cotype. It found related developments but no source stating this quotient construction or resolving the exact questions. Novelty confidence is moderate pending specialist review. The proof itself is a direct consequence of the definitions and the torus result already recorded in the source.

References

- [1] J. García-Cuerva and J. Parcet, *Quantized orthonormal systems: A non-commutative Kwapień theorem*, *Studia Math.* 155 (2003), 273–294; arXiv:math/0312245. <https://arxiv.org/abs/math/0312245>.